

Abstract

Abstract I

Abstract II

The reproducibility problem in computational research is partly structural: research artifacts are often distributed across editors, notebooks, scripts, and manual publication steps without a shared mechanism that keeps code, data, and manuscript synchronized. Existing tools address important parts of this lifecycle: workflow managers orchestrate computation, literate-programming systems render documents, and data-versioning tools track artifacts. `template/` combines those concerns with test and provenance gates by applying Infrastructure as Code to the research lifecycle. Its Two-Layer Architecture separates 28 infrastructure subdirectories (25 importable Python packages, ~662 modules, validated by ~8,583 tests) from self-contained project workspaces. A YAML-declared pipeline (16 stages; 10 in a default full run) coordinates environment checks, test execution, analysis, Pandoc/XeLaTeX rendering, SHA-256 hashing with optional steganographic watermarking, structural validation, and optional LLM review. The declared quality floors—90% for project source and 60% for infrastructure source—are injected from executable configuration rather than copied into the manuscript